10. Read Data

10.1 Experiment Objective

This course focuses on learning how to read parameters such as the robot's battery level, servo angles, and IMU data.

10.2 Experiment Preparation

This course involves the following functions from the Muto hexapod robot's Python library:

read_version(): Reads the underlying firmware version number. Returns the version number on success, or `None` on failure.

read_battery(voltage=True): Reads the battery level. The optional parameter `voltage=True` returns the voltage value, while `voltage=False` returns the battery percentage.

read_motor(): Reads the angle values of the eighteen servos. Returns an array on success, or `None` on failure.

read_leg(leg_id): Reads the angles of the servos on a specific leg. The `leg_id` parameter has a value range of [1-6], representing leg1 to leg6. Returns an array on success, or `None` on failure.

read_IMU(): Reads the fused IMU angles. Returns an array `[roll, pitch, yaw, temp]` on success, or `None` on failure.

read_IMU_Raw(): Reads the raw IMU data. Returns an array `[accX, accY, accZ, gyroX, gyroY, gyroZ, magX, magY, magZ]` on success, or `None` on failure.

10.3 Experiment Procedure

Open the JupyterLab client and navigate to the code path:

```
muto/Samples/Control/10.read_data.ipynb
```

Read and print the version number:

```
version = g_bot.read_version()
print("version:", version)
```

Read and print the battery voltage:

```
battery = g_bot.read_battery()
print("battery:%.1fV" % battery)
```

Read and print the angle values of the 18 servos. Due to factors such as pressure and conversion accuracy, the read angles may have a certain error compared to the actual angles. A deviation of ±3° from the actual angle is considered normal.

```
angle = g_bot.read_motor()
print("angle:", angle)
```

Read and print the angles of the three servos on leg 1. Due to factors such as pressure and conversion accuracy, the read angles may have a certain error compared to the actual angles. A deviation of ±3° from the actual angle is considered normal.

```
leg_id = 1
leg_angle = g_bot.read_leg(leg_id)
print("leg_angle:", leg_angle)
```

Read and print the fused IMU data:

```
imu = g_bot.read_IMU()
print("imu:", imu)
```

Read and print the raw IMU data:

```
imu_raw = g_bot.read_IMU_Raw()
print("imu_raw:", imu_raw)
```

10.4 Experiment Summary

Example of reading the battery voltage value. If you need to read the battery percentage, you can pass `False` as the parameter.

```
]: battery = g_bot.read_battery()
   print("battery:%.1fV" % battery)

   battery:7.6V
```

Example of reading the angle values of the 18 servos:

```
: angle = g_bot.read_motor()
  print("angle:", angle)

  angle: [0, -36, -20, 0, -36, -21, 0, -36, -21, -1, -36, -20, -1, -36, -20, 0, -36, -20]
```

Example of reading the fused IMU angles:

```
imu = g_bot.read_IMU()
print("imu:", imu)

imu: [-0.42, -0.47, -7.49, 20]
```